

EDUCATION

Pohang University of Science and Technology (POSTECH)

M.S. in Computer Science and Engineering, Advisor: Seungyong Lee

Mar. 2021 –Feb. 2025

Thesis : *High-Quality Geometry and Texture Editing of Neural Radiance Field*

Pohang University of Science and Technology (POSTECH)

B.S. in Computer Science and Engineering, GPA: 3.73/4.3

Mar. 2017 –Feb. 2021

EXPERIENCE

Osstem Implant

Research Intern

Winter 2024

- Propose algorithm and implemented with C++ for partial remeshing for mesh smoothing and modifications.

S-Core

Engineering Intern

Winter 2018

- Conducted big-data analysis on sales data from the company’s cafe and contributed to the development of a recommendation system.

LANGO

Website developer

Summer 2018

- Designed and developed a web page on AWS, leveraging RDS and EC2, to facilitate interactive generation and editing of XML file data.

PUBLICATIONS

1. Y. Jung, W. Jang, **S. Kim**, J. Yang, X. Tong, and S. Lee, “Deep Deformable 3D Caricatures with Learned Shape Control”, *ACM SIGGRAPH 2022 Conference Proceedings*, 2022
2. J. Kam, J. Kim, **S. Kim**, J. Park, and S. Lee, “CostDCNet: Cost Volume based Depth Completion for a Single RGB-D Image”, *Proceeding of European Conference on Computer Vision (ECCV)*, 2022
3. J. Kim, **S. Kim**, J. Park, S. Lee, “Deep Cost Ray Fusion for Sparse Depth Video Completion”, *Proceeding of European Conference on Computer Vision (ECCV)*, 2024
4. **S. Kim**, J. Son, G. Ju, J. Lee and S. Lee, “High-Quality Geometry and Texture Editing of Neural Radiance Field”, *Pacific Graphics (PG) 2024 Conference Proceedings*, 2024

DOMESTIC PUBLICATIONS

1. W. Jin, **S. Kim**, J. Park, S. Lee, and S. Cho, “GC-NeRF: Geometric Constrained NeRF for Efficient Training”, *Proceedings of 34th Workshop on Image Processing and Image Understanding (IPIU)*, 2022.
2. **S. Kim**, and S. Lee, “Texture Reconstruction for View-dependent Colors”, *Proceedings of the Korea Computer Graphics Society Conference*, 2022.
3. **S. Kim**, J. Son, G. Ju, J. Lee and S. Lee, “*uvh*-TensorRF: High-quality Mesh-based Neural Rendering”, *Proceedings of 34th Workshop on Image Processing and Image Understanding (IPIU)*, 2024.

TEACHING

- **Teaching Assistant** at Pohang University of Science and Technology (POSTECH) Fall 2022
Algorithms and Problem Solving Practice (CSED490E)
- **Teaching Assistant** at Pohang University of Science and Technology (POSTECH) Spring 2022
Computer Graphics (CSED451)

PROJECTS

- Neural Texture for View-dependent Colors 2022
Research project and academic publication.
Neural texture and MLP for real-time novel view synthesis.
- NeRF Training with Geometric Loss 2022
Research project and academic publication.
Designing additional losses based on reconstructed meshes from input images to improve NeRF training.
- Google QUEST QA Labeling Challenge 2020
Kaggle code competition. (127th place out of 1571 teams, top 9%)
Analyzing BERT language model, model search and hyperparameter tuning.
- Next Best View Determination 2020
Research project.
Developing a novel segmentation algorithm for mesh saliency calculation to detect large noisy regions.

SCHOLARSHIPS AND AWARDS

- PPC (POSTECH Programming Contest) 3rd place winner 2018
- Samsung SW algorithm competition 3rd place winner 2018
- POSTECH Global Leadership Program (Scholarship Program) 2018

SKILLS

- **Programming Language:** C, C++, Python, Java

LANGUAGES

- **Korean:** Native Speaker
- **Japanese:** Bilingual Proficiency
- **English:** Intermediate

EXTRACURRICULAR ACTIVITIES

- SIGGRAPH Asia Student Volunteer (SV) 2022
Volunteering at SIGGRAPH Asia 2022 in Daegu.
- POSTECH Broadcasting System 2017–2019
Program director and video producer